

# Numerical Analysis PhD Exam, May 2014

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*You have 2 hours to answer the following questions. You must show all your work as neatly and clearly as possible and indicate the final answer clearly. You may not use a calculator.*

1. Show that the fixed point equation  $x = f(x)$  with a fixed point  $\alpha$  can be solved iteratively, if  $f'(\alpha) \neq 1$ , by one of the following fixed point iteration formulas:

(a)  $x_{n+1} = f(x_n)$

(b)  $x_{n+1} = f^{-1}(x_n)$ . Hint: The following formula for the derivative of an inverse is valid

$$[f^{-1}]'(x) = \frac{1}{f'(f^{-1}(x))}.$$

(c)  $x_{n+1} = (x_n + f(x_n))/2$

2. Let  $f(x) \in C^1[a, b]$ . Let  $p(x)$  be a polynomial for which

$$\|f' - p\|_{\infty} \leq \epsilon$$

and define

$$q(x) = f(a) + \int_a^x p(t) dt, \quad a \leq x \leq b.$$

Show that  $q(x)$  is a polynomial that satisfies

$$\|f - q\|_{\infty} \leq \epsilon(b - a).$$

3. Consider the integral

$$\int_0^{\pi} x^2 \cos x dx. \tag{1}$$

- (a) Consider a quadrature rule of the form

$$\int_0^{\pi} x^{\alpha} f(x) dx \approx Af(0) + B \int_0^{\pi} f(x) dx$$

where  $\alpha > -1$ ,  $\alpha \neq 0$  is a parameter. Determine the constants  $A, B$  so that the quadrature formula has degree of exactness one.

- (b) Use the formula in part (a) to approximate the integral (1).

4. For the function  $f(x) = \ln(1 + x)$  for  $x \in [0, 1]$ , find the minimax approximation polynomial of degree one. Give the exact value of the minimax error.

5. Assume that you are solving the initial value problem

$$\begin{aligned} y' &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= \alpha. \end{aligned}$$

- (a) Derive the formula for the global error of the numerical solutions for the ODE problem above obtained via Euler's method. Hint: The formula is  $M = \|Y''\|_\infty$ :

$$|Y(t_i) - w_i| < \frac{hM}{2L} [e^{L(b-a)} - 1] .$$

- (b) Compute the value of  $M = \|Y''\|_\infty$  necessary to apply the global error formula above to the specific ODE problem

$$\begin{aligned} y' &= \sin(t + 2y) + e^t, \quad 0 \leq t \leq 1, \\ y(0) &= 0. \end{aligned}$$